

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 19: Earth, Moon and the Sun

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	19 — Earth, Moon and the Sun
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct.

Traps: phases are NOT Earth's shadow (that is a lunar eclipse); the Moon's shape does not really change — only the lit fraction we see does; the far side is not permanently dark; same-face comes from rotation EQUALLING revolution (not from no spin); reflected light, not self-made light; surface conditions follow from no air / no liquid water / low gravity; day–night is rotation geometry, not the Sun moving or switching off. Do not write on this paper — mark answers on your answer sheet.

1. If the Sun were suddenly hidden behind a giant screen so that no sunlight could reach the Moon, what would happen to the Moon as seen from Earth?

- (A) It would keep glowing, because the Moon stores heat from the day and releases it as light at night.
- (B) It would still shine faintly, because the Moon makes a little of its own light like a small star.
- (C) It would turn red, because Earth's atmosphere would light it up instead of the Sun.
- (D) It would become invisible, because the Moon has no light of its own and only shines by reflecting sunlight.

2. A new Moon is nearly invisible from Earth. The best reason is that at new Moon:

- (A) the Moon has moved into Earth's shadow and gets no sunlight at all.
- (B) the Moon temporarily stops reflecting light for a night.
- (C) the Moon shrinks to its smallest real size during that part of the month.
- (D) the Moon's sunlit half is turned away from Earth, so the side facing us is unlit.

3. The Moon's phases repeat in the same order every month. Which sequence is correct going forward in time?

- (A) new Moon → waxing gibbous → first quarter → waxing crescent → full Moon
- (B) new Moon → waxing crescent → first quarter → waxing gibbous → full Moon
- (C) full Moon → waxing crescent → first quarter → new Moon → waxing gibbous
- (D) first quarter → new Moon → waxing crescent → full Moon → waxing gibbous

4. A student insists the Moon's phases are caused by Earth's shadow falling on the Moon. The strongest single objection is:

- (A) Earth's shadow is too small to ever reach the Moon, so it can never darken it.
- (B) the Moon is too bright for any shadow to affect what we see.
- (C) phases are really caused by clouds in Earth's atmosphere covering parts of the Moon.
- (D) Earth's shadow on the Moon produces a lunar eclipse, which is rare — not the smooth set of phases we see every month.

5. We always see the same face of the Moon because the Moon:

- (A) does not rotate on its axis at all.
- (B) rotates once on its axis in exactly the same time it takes to orbit Earth once.
- (C) spins so fast that the far side blurs out of view.
- (D) is held perfectly still by Earth's magnetic field.

6. Suppose the Moon orbited Earth exactly as it does now but did NOT rotate on its axis at all. Over one full orbit, an observer on Earth would:

- (A) see exactly the same single face the whole time, just as we do now.
- (B) see the Moon disappear, because a non-spinning Moon cannot reflect light.
- (C) see the Moon flip upside-down once each night.
- (D) get to see all of the Moon's surface, because its face would no longer stay pointed at us.

7. There is complete silence on the Moon — a hammer blow makes no sound that another astronaut nearby can hear. The reason is that the Moon has:

- (A) too strong a gravity, which crushes sound waves before they spread.
- (B) a surface so cold that sound freezes instantly.
- (C) no atmosphere, and sound needs a medium such as air to travel through.
- (D) a fast spin that scatters the sound away.

8. Footprints left by astronauts on the Moon in 1969 are expected to stay almost unchanged for millions of years. This is mainly because the Moon has:

- (A) no wind and no liquid water, so there is nothing to erode or wash the prints away.
- (B) such strong gravity that the dust is pressed permanently flat.
- (C) a thick atmosphere that seals the prints under air pressure.
- (D) constant volcanic activity that re-bakes the surface and hardens it.

9. Daytime temperatures on the Moon's surface can soar well above boiling water while night-time temperatures plunge far below freezing. The best explanation for this huge swing is that the Moon:

- (A) is much closer to the Sun than Earth is, so it overheats by day.
- (B) has no atmosphere to trap and spread heat, so the surface heats and cools sharply.
- (C) is made of materials that create their own heat in sunlight.
- (D) spins so fast that one side bakes while the other freezes.

10. On the Moon an astronaut can jump far higher than on Earth using the same effort. This is because the Moon's surface gravity is about:

- (A) the same as Earth's, but the spacesuit is lighter there.
- (B) one-sixth of Earth's, so each jump lifts the astronaut much higher.
- (C) six times Earth's, which springs the astronaut upward.
- (D) exactly zero, so the astronaut floats freely.

11. When you see a thin crescent Moon, the curved bright edge and the two horns always point away from a particular direction. The bright side faces:

- (A) toward Earth's shadow, which carves out the dark part.
- (B) toward the Sun, because that is the direction the sunlight is coming from.
- (C) in a random direction that changes minute to minute.
- (D) always straight up, regardless of where the Sun is.

12. A 'first-quarter' Moon looks like a half-lit disc in the sky, yet it is called a quarter. This name makes sense because at that point the Moon has:

- (A) completed about one quarter of its monthly orbit around Earth since new Moon.
- (B) exactly one quarter of its surface lit by the Sun at that moment.
- (C) shrunk to a quarter of its real size for that part of the month.
- (D) moved a quarter of the way into Earth's shadow.

13. Two astronauts stand on the Moon's far side — the side that never faces Earth. During the lunar day there, the far side is:

- (A) brightly lit by the Sun, even though it can never be seen from Earth.
- (B) always in total darkness, which is why it is the permanent 'dark side'.
- (C) lit only by light bounced from Earth, never by the Sun directly.
- (D) frozen solid forever because no sunlight ever reaches it.

14. It is daytime in India while it is night in Brazil at the same instant. The reason both happen together is that:

- (A) the Sun shines on the whole Earth at once, but clouds block it over Brazil.
- (B) the Sun switches off over Brazil while it stays on over India.
- (C) the Sun lights only the half of Earth facing it, and rotation carries each place through the lit and unlit halves.
- (D) Earth's shadow from the Moon covers Brazil at that moment.

15. From India the Sun appears to rise in the east and set in the west. This apparent motion is really produced by:

- (A) the Sun physically orbiting the Earth once every day.
- (B) the Sun being dragged across the sky by the Moon's gravity.
- (C) Earth revolving around the Sun once each day.
- (D) Earth spinning from west to east, which makes the Sun seem to drift the opposite way across the sky.

16. The Sun is the chief source of light and heat for Earth, while the night sky's other stars give us almost no warmth. The main reason the Sun is so much more powerful for us is that:

- (A) it is the largest and hottest star in the entire universe.
- (B) it is a star far, far closer to Earth than any other star.
- (C) it is the only object in space that produces light and heat.
- (D) it is a planet that has been set on fire to warm Earth.

17. A full Moon and a new Moon are exactly opposite phases. Which statement correctly contrasts them?

- (A) At full Moon the Sun, Earth, and Moon are nearly in line with Earth in the middle; at new Moon the Moon is between Earth and the Sun.
- (B) At full Moon the Moon is between Earth and the Sun; at new Moon Earth is in the middle.
- (C) Both occur when the Moon sits exactly inside Earth's shadow.
- (D) At full Moon the Moon makes its own light, while at new Moon it only reflects sunlight.

18. The Moon takes about 29.5 days between one full Moon and the next, yet it actually completes a full circle against the background stars in about 27.3 days. The extra ~2 days are needed because:

- (A) the Moon slows down near full Moon and speeds up near new Moon.
- (B) Earth's shadow delays the Moon's return to full each month.
- (C) while the Moon orbits, Earth has also moved along its own orbit around the Sun, so the Moon must travel a little farther to line up with the Sun again.
- (D) the Moon stops for about two days at full Moon before continuing.

19. A waning gibbous Moon is seen late at night. 'Waning' means that night after night the lit part is:

- (A) shrinking, on its way from full Moon back toward new Moon.
- (B) growing, on its way from new Moon toward full Moon.
- (C) staying exactly the same until the next new Moon.
- (D) flickering larger and smaller from one hour to the next.

20. Even though the Sun is about 400 times wider than the Moon, both look roughly the same size in our sky. The explanation is that:

- (A) the Sun is about 400 times farther from Earth than the Moon, so the huge size and huge distance cancel out.
- (B) the Moon is secretly almost as big as the Sun.
- (C) the Sun shrinks when it is high in the sky.
- (D) the Moon magnifies itself by reflecting sunlight.

21. An astronaut on the Moon looks back at Earth and sees it going through phases too — a 'crescent Earth', a 'full Earth', and so on. The reason Earth shows phases to a Moon observer is that:

- (A) Earth makes its own light that brightens and dims over the month.
- (B) the Moon's shadow falls across Earth and carves out the phases.
- (C) the astronaut sees varying amounts of Earth's sunlit half, just as we on Earth see varying amounts of the Moon's sunlit half.
- (D) Earth physically changes shape as the Moon orbits it.

22. The boundary line on the Moon between its lit part and its dark part — the terminator — is where:

- (A) Earth's shadow ends and full sunlight begins.
- (B) the Moon's surface is permanently split into a hot half and a cold half forever.
- (C) the Sun is just rising or setting for that part of the Moon's surface.
- (D) the Moon's own light fades into darkness.

23. Why can a solar eclipse happen only at new Moon and a lunar eclipse only at full Moon?

- (A) Because the Moon makes extra light at those phases that triggers an eclipse.
- (B) Only at those phases can the Sun, Moon, and Earth line up closely enough for one to cast its shadow on another.
- (C) Because the Moon is closest to Earth only at new and full Moon.
- (D) Because Earth stops rotating at new and full Moon.

24. During a total lunar eclipse the fully shadowed Moon often glows a dull coppery red rather than going completely black. This happens because Earth's atmosphere:

- (A) produces its own red light that shines onto the Moon.
- (B) bends some red sunlight around the edge of Earth and into its shadow, where it falls on the Moon.
- (C) sets fire to the Moon's surface, making it glow red.
- (D) reflects the red colour of Earth's oceans up to the Moon.

25. A child reasons: 'The Moon must be a star, because both shine in the night sky.' The clearest flaw in this reasoning is that:

- (A) the Moon is hotter than any star, so it cannot be a star.
- (B) stars only appear in the daytime, never at night.
- (C) a star produces its own light, while the Moon only reflects light from the Sun.
- (D) the Moon is larger than every star, so it must be something else.

26. Tonight the Moon is a thin crescent. Roughly a week later, if the lit part has grown to a half-disc, the Moon has reached the:

- (A) full Moon phase.
- (B) new Moon phase.
- (C) waning crescent phase.
- (D) first-quarter phase.

27. Compared with Earth, the Moon, and the Sun, the correct ranking from largest to smallest in actual size is:

- (A) Earth, then Sun, then Moon.
- (B) Sun, then Moon, then Earth.
- (C) Moon, then Earth, then Sun.
- (D) Sun, then Earth, then Moon.

28. The Moon's craters, formed by impacts billions of years ago, are still sharp and well preserved today. On Earth most ancient craters have been worn away. The key difference is that:

- (A) the Moon has no air and no liquid water, so weathering and erosion barely act on it.
- (B) the Moon's craters are far younger than any on Earth.
- (C) the Moon constantly forms fresh craters that look old.
- (D) the Moon's strong gravity protects the craters from wearing down.

29. If you watch the Moon over a single night, it appears to drift slowly across the sky from east to west, much as the Sun does by day. This nightly east-to-west drift is mainly caused by:

- (A) the Moon racing around Earth once every night.
- (B) the Moon being pushed westward by the Sun's light.
- (C) Earth orbiting the Sun once each night.
- (D) Earth's rotation, which makes all sky objects appear to move from east to west.

30. A waxing crescent Moon is usually seen low in the western sky just after sunset, not in the middle of the night. This timing fits the geometry because a thin waxing crescent lies:

- (A) directly opposite the Sun, so it rises just as the Sun sets.
- (B) exactly overhead at midnight every night of its phase.
- (C) close to the Sun's direction in the sky, so it appears near the Sun and sets soon after it.
- (D) inside Earth's shadow, which is why only a sliver shows.

31. Astronauts who visited the Moon found no flowing rivers, no rain, and no oceans. The fundamental reason the Moon has no liquid water on its surface is that:

- (A) the Moon is too close to the Sun, so all its water boiled away into the Sun.
- (B) the Moon's strong gravity pulls all water deep underground forever.

(C) the Moon never had any water-making materials in the first place, so liquid is impossible.

(D) with no atmosphere and extreme temperatures, water cannot stay liquid on the open surface.

32. At the exact moment of a full Moon, the Moon rises in the east just as the Sun sets in the west. This neat timing follows from the fact that at full Moon:

(A) the Moon is between Earth and the Sun, so they share the same part of the sky.

(B) the Moon is on the opposite side of Earth from the Sun, so as the Sun goes down the Moon comes up.

(C) the Moon makes extra light that lets it rise early.

(D) Earth stops spinning briefly, letting the Moon appear.

33. From Earth we see the Moon shine, yet half of the Moon is always in sunlight. How can both be true?

(A) The Sun always lights half the Moon; from Earth we see only part of that lit half, and that visible fraction is what changes through the phases.

(B) Only the part we can see from Earth is ever lit by the Sun.

(C) The whole Moon is lit at full Moon and the whole Moon is dark at new Moon.

(D) The Moon lights its own visible half and leaves the other half dark.

34. A spacecraft can orbit the Moon and photograph its far side, but a person standing on Earth can never see that side. The reason is that:

(A) the far side is always dark, so a camera on Earth gets no image of it.

(B) the Moon's spin keeps the same face toward Earth, so the far side never turns to face us.

(C) the far side is hidden inside Earth's permanent shadow.

(D) the far side faces away from the Sun at all times and so is invisible.

35. Earth completes one rotation on its axis in about a day, while the Moon completes one rotation in about a month. Because of this difference, a single 'day–night cycle' on the Moon (one sunrise to the next) lasts about:

(A) 24 hours, the same as a day on Earth.

(B) one hour, because the Moon is small and spins fast.

(C) a month, since the Moon turns much more slowly than Earth.

(D) a year, the same as Earth's trip around the Sun.

36. Earthshine is the faint glow that lets us dimly see the whole disc of the Moon during a thin crescent — the dark part is softly lit. The light making that dark part visible is:

(A) light the Moon makes on its own in its dark regions.

(B) sunlight reflected off Earth onto the Moon, then reflected back to us.

(C) heat glowing out of the Moon's interior.

(D) light from distant stars collecting on the Moon's dark side.

37. If Earth rotated half as fast as it does now (one spin taking about 48 hours), the most direct effect would be that:

- (A) the year would become twice as long.
- (B) days and nights would each last about twice as long.
- (C) the seasons would disappear entirely.
- (D) the Sun would stop rising in the east.

38. A 'waxing gibbous' Moon shows more than half its disc lit and is growing toward full. In the Sun–Earth–Moon geometry, a gibbous Moon occurs when the Moon is:

- (A) more than a quarter-turn but less than half-turn around its orbit, so we see most of the lit half.
- (B) directly between Earth and the Sun, hiding its lit side.
- (C) exactly between the new and crescent positions.
- (D) passing through the middle of Earth's shadow.

39. On Earth we feel warmth from sunlight directly and also benefit from moonlight at night. A correct comparison of the two is:

- (A) sunlight is energy made by the Sun itself; moonlight is just sunlight reflected off the Moon, far weaker.
- (B) both sunlight and moonlight are made independently by the Sun and the Moon.
- (C) moonlight is stronger than sunlight because the Moon is closer.
- (D) moonlight is heat stored by the Moon during the day and released at night.

40. Suppose, impossibly, that the Moon stopped orbiting Earth and just hung still in space at a fixed point, while the Sun and Earth kept moving normally. Over a month, an Earth observer watching the Moon would notice that:

- (A) the Moon would freeze in one single phase forever.
- (B) the Moon would stop reflecting light and vanish.
- (C) the Moon would start making its own light to compensate.
- (D) the Moon would still go through phases, because the Sun-Moon-Earth angle would keep changing as Earth moves.

41. Why does the Sun look like a large, bright disc while other stars look like tiny twinkling points, even though many of those stars are bigger than the Sun?

- (A) The Sun is the only true star; the others are just reflective rocks.
- (B) The Sun is the biggest star in existence, dwarfing all others.
- (C) The Sun is enormously closer to Earth than the other stars, so it looks large while they look like points.
- (D) Other stars are dim because they make no light of their own.

42. Imagine standing at Earth's North Pole during the long polar summer, when the Sun stays above the horizon for weeks. This continuous daylight happens because:

- (A) Earth's tilt keeps the North Pole leaning toward the Sun, so rotation never carries it into the dark half.
- (B) Earth stops rotating at the poles during summer.
- (C) the Sun grows large enough to light the whole Earth at once.
- (D) the Moon reflects extra sunlight onto the pole all night.

43. A student says the Moon must rotate because we sometimes see slightly different edges of it over a month. In fact, the steady, basic reason the Moon shows essentially one face is that its:

- (A) rotation is far faster than its revolution, blurring the far side away.
- (B) revolution is far faster than its rotation, so the same side leads.
- (C) rotation period and revolution period are equal, so it turns just enough to keep the same face toward us.
- (D) rotation has completely stopped, leaving one face frozen at us.

44. On Earth, dropping a feather and a hammer, the hammer falls faster because of air. Astronauts on the Moon famously dropped them and they hit the ground together. The reason is that the Moon:

- (A) has gravity so strong it forces both to fall at the same instant.
- (B) has zero gravity, so both simply float down together.
- (C) has no air, so there is no air resistance to slow the lighter feather.
- (D) has a magnetic field that pulls metal and feather equally.

45. Between successive new Moons, the Moon passes through first quarter and full Moon. Which best describes how the lit portion we see changes over this whole cycle?

- (A) It grows steadily from new Moon all the way to the next new Moon without ever shrinking.
- (B) It stays fully lit the entire cycle and only the Moon's position changes.
- (C) It grows from nothing at new Moon to fully lit at full Moon, then shrinks back to nothing at the next new Moon.
- (D) It jumps suddenly from dark to fully lit at midnight on the full-Moon date.

46. The Sun is described in modern astronomy as 'an ordinary, average star.' Calling it ordinary makes sense because:

- (A) it is just one of countless similar stars; it only seems special to us because it is so close.
- (B) it is the smallest and dimmest star known.
- (C) it is unique and unlike any other object in the universe.
- (D) it is actually a large planet rather than a star.

47. On the Moon a coin and a sheet of paper, released from the same height, land at the same time. If the same experiment is done in a sealed room on Earth that still has air, the paper lands later. This contrast shows that:

- (A) gravity is stronger on the Moon, equalising the fall.
- (B) air resistance, present on Earth but absent on the Moon, is what slows the paper.
- (C) the Moon has no gravity, so timing is meaningless there.
- (D) paper is heavier on the Moon than a coin is.

48. It is easy to look straight at the full Moon but dangerous to stare at the Sun. The basic difference behind this is that:

- (A) the Moon is actually brighter but our eyes ignore it at night.
- (B) the Sun reflects light from the Moon, concentrating it.
- (C) the Moon absorbs sunlight, so it gives off no light at all.
- (D) the Sun produces its own intense light, while the Moon only sends back a small fraction of reflected sunlight.

49. From a spaceship far out in space, at any single instant Earth appears half lit and half dark by the Sun. As Earth spins, a city on its surface experiences day and night because:

- (A) the Sun moves around to light first one city and then another.
- (B) the city's own lights switch on and off the surface glow.
- (C) the Moon's shadow sweeps across each city in turn.
- (D) the city is carried from the lit half into the dark half and back as Earth rotates.

50. A 'last-quarter' (third-quarter) Moon shows a half-disc lit on the opposite side from a first-quarter Moon. In the monthly cycle the last quarter comes:

- (A) before first quarter, while the lit part is still waxing toward full.
- (B) exactly at the moment of new Moon, when no light shows.
- (C) after full Moon, while the lit part is waning toward the next new Moon.
- (D) only during a lunar eclipse, when Earth's shadow halves the Moon.

51. Why do meteoroids that would burn up as bright 'shooting stars' in Earth's sky simply slam into the Moon and leave craters instead?

- (A) The Moon's gravity is too weak to attract any meteoroids.
- (B) The Moon has no atmosphere to make them heat up and burn, so they strike the surface directly.
- (C) The Moon is too cold for meteoroids to burn.
- (D) The Moon reflects meteoroids away before they can land.

52. Earth turns from west to east. Because of this spin direction, for an observer in India the Sun, Moon, and stars all appear to:

- (A) move across the sky from east to west.
- (B) move across the sky from west to east, the same way Earth turns.
- (C) stay fixed in place while only the ground moves.
- (D) rise in the north and set in the south.

53. Two friends argue about the Moon's 'dark side.' One says it is permanently dark; the other says it gets sunlight. Who is right and why?

- (A) The second friend: every part of the Moon, including the far side, gets sunlight as the Moon rotates; it is just never visible from Earth.
- (B) The first friend: the far side never receives any sunlight at all.
- (C) Neither: the Moon has no far side because it does not rotate.
- (D) The first friend: the far side is locked inside Earth's shadow.

54. A learner thinks every full Moon should produce a lunar eclipse, since Earth is then between the Sun and Moon. Why are lunar eclipses actually much rarer than full Moons?

- (A) Earth's shadow is far too small to ever reach the Moon's orbit.
- (B) Lunar eclipses need a new Moon, not a full Moon.
- (C) The Moon must make its own light for an eclipse, which it rarely does.
- (D) The Moon's orbit is slightly tilted, so usually at full Moon the Moon passes a little above or below Earth's shadow rather than into it.

55. On the Moon there is no twilight: it goes from full daylight to darkness almost instantly as the Sun dips below the horizon, unlike Earth's gradual dusk. The reason is that the Moon has:

- (A) no atmosphere to scatter sunlight after the Sun sets, so there is no lingering glow.
- (B) a very thick atmosphere that blocks the Sun all at once.
- (C) such fast rotation that the Sun seems to vanish in seconds.
- (D) its own light that it switches off at sunset.

56. The phrase 'the Moon shines' is loose. The most scientifically accurate rewording is:

- (A) the Moon burns gases to produce its own glow.
- (B) the Moon reflects sunlight toward us, so it appears bright.
- (C) the Moon gives off light stored from earlier sunlight.
- (D) the Moon glows because it is extremely hot inside.

57. Place these in order of increasing distance from Earth: the Moon, the Sun, and the next-nearest star after the Sun. The correct order (nearest first) is:

- (A) the Sun, then the Moon, then the next-nearest star.
- (B) the next-nearest star, then the Sun, then the Moon.
- (C) the Moon, then the next-nearest star, then the Sun.
- (D) the Moon, then the Sun, then the next-nearest star.

58. If Earth were tidally locked to the Sun the way the Moon is to Earth — always keeping one face toward the Sun — the most direct consequence would be that:

- (A) Earth would have no seasons but normal days and nights.
- (B) the whole Earth would be lit at once with no night anywhere.
- (C) one side of Earth would have permanent day and the other permanent night.
- (D) Earth would stop orbiting the Sun entirely.

59. Looking at a waxing crescent and a waning crescent, both are thin slivers, but the lit edge is on opposite sides. The single feature that tells them apart over several nights is that:

- (A) the waxing crescent is hotter, while the waning crescent is colder.
- (B) the waxing crescent makes its own light, while the waning crescent reflects sunlight.
- (C) the waxing crescent grows fatter night after night, while the waning crescent grows thinner.
- (D) the waxing crescent appears only at the North Pole and the waning crescent only at the South Pole.

60. An astronaut standing on the Moon at noon sees the sky around the Sun as black, not blue like Earth's daytime sky, even though the Sun is shining brightly. The reason is that the Moon has:

- (A) a dark-coloured atmosphere that absorbs all blue light.
- (B) moved into Earth's shadow, which blackens the whole sky.
- (C) no sunlight reaching it at noon, so the sky must be dark.
- (D) no atmosphere to scatter sunlight, so apart from the Sun the sky stays black.

Solutions — Science (Class 7), Chapter 19: Earth, Moon and the Sun — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	C	31	D	41	C	51	B
2	D	12	A	22	C	32	B	42	A	52	A
3	B	13	A	23	B	33	A	43	C	53	A
4	D	14	C	24	B	34	B	44	C	54	D
5	B	15	D	25	C	35	C	45	C	55	A
6	D	16	B	26	D	36	B	46	A	56	B
7	C	17	A	27	D	37	B	47	B	57	D
8	A	18	C	28	A	38	A	48	D	58	C
9	B	19	A	29	D	39	A	49	D	59	C
10	B	20	A	30	C	40	D	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) The Moon is not a light source; everything we see of it is sunlight bounced off its surface. Cut off the sunlight and there is nothing to reflect, so the Moon would vanish from view. It does not store and re-emit light, it produces no light of its own (unlike a star which makes light by fusion), and there is no Earth-glow strong enough to make it shine on its own — those options assume the Moon is a source of light, which it is not.

2. (D) At new Moon the Moon lies roughly between Earth and the Sun, so its lit half points toward the Sun and away from us; the near side is dark, making it nearly invisible. It is NOT in Earth's shadow (that situation is a lunar eclipse, which happens at full Moon), the Moon never stops reflecting, and the Moon does not actually change size — only the visible lit fraction changes.

3. (B) After new Moon the lit part we see grows (waxes): a thin crescent first, then half-lit at first quarter, then more than half (gibbous), then fully lit at full Moon. The correct order is new → waxing crescent → first quarter → waxing gibbous → full. The other orders

jumble the growing sequence — gibbous (more than half) cannot come before the crescent (less than half), and the lit fraction must increase steadily from new to full.

4. (D) Phases come from seeing different amounts of the Moon's sunlit half as it orbits Earth. Earth's shadow does fall on the Moon occasionally — but that event is a lunar eclipse, which happens only a couple of times a year, not the steady monthly cycle of phases. Earth's shadow can reach the Moon (that is why lunar eclipses exist), brightness is irrelevant to the shape, and clouds cannot create the precise repeating phase pattern.

5. (B) If the Moon did not spin at all, we would actually get to see all its sides over one orbit. Because its spin period equals its orbit period (about 27.3 days each), it turns to keep the same face pointed at Earth the whole way around. A fast spin would show us all sides too, and Earth's magnetism is far too weak to lock a body the size of the Moon.

6. (D) Keeping one face toward Earth actually requires the Moon to turn once per orbit. A Moon that did not spin would keep its face pointed at the same distant direction in space while it circled us, so over one orbit Earth would face every side of it in turn. Reflection does not depend on spin, and the flipping-each-night idea confuses the daily rotation of Earth with the Moon's monthly motion.

7. (C) Sound is a vibration that must pass through a material like air, water, or rock to travel; with no air on the Moon, vibrations cannot reach a listener through the empty space between them. (Vibrations could still pass through the solid ground.) Gravity does not block sound, cold does not 'freeze' it, and the Moon's slow spin is irrelevant — the missing ingredient is air.

8. (A) On Earth wind and flowing water quickly wipe out footprints. The Moon has no atmosphere (no wind) and no liquid water, so the usual agents of erosion are absent and prints persist for ages. The Moon's gravity is weak, not strong; it has essentially no atmosphere rather than a thick one; and it is not volcanically active in a way that would resurface footprints.

9. (B) On Earth the atmosphere acts like a blanket, storing daytime heat and moderating the night, so temperatures stay in a narrow range. The Moon has no such blanket, so sunlit ground gets very hot and shaded ground gets very cold. The Moon is essentially the same distance from the Sun as Earth, it does not generate heat by itself, and its slow spin actually gives long hot days and long cold nights — but the root cause is the missing atmosphere.

10. (B) The Moon's gravity is about one-sixth of Earth's, so an astronaut weighs about a sixth as much and the same push sends them roughly six times higher. Gravity there is not the same as Earth's, not six times stronger (which would make jumping harder), and not zero — they still fall back down, just slowly.

11. (B) The lit part of the Moon is the half being struck by sunlight, so the bright curve always bulges toward the Sun and the horns of the crescent point away from it. The dark part is simply the unlit far portion, not Earth's shadow; the direction is set by the Sun, not random; and it is not fixed straight up — it tilts to face wherever the Sun is.

12. (A) The 'quarter' refers to the fraction of the orbit completed, not the fraction we see lit: about a quarter of the ~29.5-day cycle has passed since new Moon. (The Sun lights half the Moon at all times; we just happen to see half of that lit half — a 'half Moon' shape.) The Moon does not change size, and Earth's shadow is not involved in phases.

13. (A) Every part of the Moon, including the far side, receives sunlight as the Moon rotates — the far side simply can never be observed from Earth because of the Moon's locked rotation. Calling it the 'dark side' is misleading: it is the 'far' side, not a permanently unlit one. It is lit directly by the Sun, not just by faint earthshine, and it is not eternally frozen.

14. (C) At any instant the Sun can only light the hemisphere pointed toward it; the other hemisphere is in Earth's own night side. As Earth rotates, places move from the lit half to the dark half, so somewhere is always in day and somewhere always in night. The Sun cannot light all of a sphere at once, it never switches off, and the Moon's shadow is tiny and rare — it cannot blanket a whole country's night.

15. (D) Earth rotates eastward (west to east), so to us the whole sky — including the Sun — seems to slide the other way, rising in the east and setting in the west. The Sun does not orbit Earth, the Moon does not drag the Sun, and Earth's revolution takes a year, not a day, so it cannot explain the daily east-to-west march of the Sun.

16. (B) The Sun is an ordinary, fairly average star, but it is enormously closer to us than any other star, so its light and heat reach us strongly while distant stars appear as faint points. It is not the biggest or hottest star, it is not the only light-and-heat source (every star makes its own), and it is a star, not a burning planet.

17. (A) At full Moon, Earth lies between the Sun and Moon, so the Moon's whole near side is sunlit; at new Moon the Moon lies between Earth and the Sun, so its lit side faces away from us. The first distractor swaps the two arrangements, Earth's shadow only touches the Moon at a lunar eclipse (not every full and new Moon), and the Moon always only reflects — it never makes its own light at any phase.

18. (C) 27.3 days is the time to circle once relative to the distant stars. But Earth keeps moving around the Sun, so after 27.3 days the Sun–Earth–Moon line is not yet repeated; the Moon needs about two more days of orbiting to catch up and reach full Moon again. The Moon does not stop or change speed for this reason, and Earth's shadow has nothing to do with the length of the phase cycle.

19. (A) 'Waning' means decreasing: after full Moon the lit fraction we see shrinks each night through waning gibbous, last quarter, and waning crescent until new Moon. 'Waxing' would mean growing (new toward full). The lit fraction changes gradually over days, not hour to hour, and it certainly does not stay frozen.

20. (A) A big object far away can look the same size as a small object close up. The Sun is about 400 times wider but also about 400 times farther, so the two appear nearly equal in the sky — which is why the Moon can just cover the Sun during a solar eclipse. The Moon is genuinely much smaller, the Sun does not change size, and reflection does not magnify an object.

21. (C) Phases are simply about how much of a sunlit sphere a viewer can see. From the Moon, Earth is half-lit by the Sun, and depending on the geometry the astronaut sees more or less of that lit half — Earth's phases. Earth does not make its own visible light, the Moon's shadow on Earth is tiny (a solar eclipse), and Earth does not change shape.

22. (C) The terminator is the line dividing day from night on the Moon — exactly where the Sun sits on the horizon (rising or setting) for the ground there. It moves as the Moon rotates. It is not the edge of Earth's shadow (that only appears in a lunar eclipse), it is not a permanent fixed split, and the Moon has no light of its own to fade.

23. (B) A solar eclipse needs the Moon between Sun and Earth (the new-Moon geometry); a lunar eclipse needs Earth between Sun and Moon (the full-Moon geometry). Only at these phases is a near-straight Sun–Moon–Earth line possible. The Moon never makes its own light, its distance from Earth is not set by phase, and Earth never stops rotating.

24. (B) Even inside Earth's shadow, sunlight passing through the thin ring of Earth's atmosphere is bent (refracted) and reddened, then reaches the Moon, giving it a coppery glow. The atmosphere does not make its own light, the Moon is not on fire, and ocean colour is not what is reaching the Moon — it is refracted, reddened sunlight.

25. (C) Stars generate their own light (the Sun is a star); the Moon makes none and is visible only because it reflects sunlight. So 'they both shine' hides a key difference. The Moon is far cooler than a star, stars are seen mostly at night (the Sun, a star, rules the day), and the Moon is vastly smaller than stars — so those other statements are false.

26. (D) After a waxing crescent the lit part keeps growing; about a week into the cycle it reaches half-lit, which is first quarter. Full Moon comes about two weeks after new Moon (not one), new Moon is when the lit side faces away (the start, not a week later), and a waning crescent is a shrinking sliver near the end of the cycle, not a growing half-disc.

27. (D) The Sun is by far the largest — a star vastly bigger than Earth. Earth is larger than the Moon (the Moon is roughly a quarter of Earth's width). So the true order is Sun > Earth > Moon. The other orderings either shrink the Sun below Earth or place the Moon above Earth, both of which are wrong.

28. (A) On Earth, wind, rain, rivers, and living things erase craters over time. The Moon lacks atmosphere and liquid water, so erosion is almost absent and old craters survive nearly unchanged. The lunar craters are ancient (not young), they are not freshly remade to look old, and the Moon's gravity is weak — it is the absence of erosion, not strong gravity, that preserves them.

29. (D) Earth spins eastward, so the Sun, Moon, and stars all appear to swing from east to west each day and night — that is what makes the Moon drift across the sky in one night. The Moon takes about a month, not a night, to orbit Earth; sunlight does not push it westward; and Earth's orbit around the Sun takes a year. The nightly drift is the spin of Earth.

30. (C) A thin crescent occurs when the Moon is angularly near the Sun (only a little of the lit half shows). Being near the Sun, it appears close to it in the sky and follows it down, visible briefly after sunset in the west. The full Moon (not the crescent) sits opposite the Sun and rises at sunset; the crescent is not overhead at midnight; and the sliver is from geometry, not Earth's shadow.

31. (D) Liquid water needs the right pressure and temperature; with essentially no atmosphere and huge temperature extremes, water on the Moon's surface cannot remain liquid (it would freeze or escape as vapour). The Moon is at about Earth's distance from the Sun (not specially close), its gravity is weak (not strong), and the claim it lacks any water-forming material is not the reasoned cause — the missing atmosphere and extreme conditions are.

32. (B) Full Moon means Earth is between the Sun and Moon, so the Moon sits opposite the Sun in our sky; when the Sun sets in the west the Moon must be rising in the east. A Moon between Earth and Sun would be a new Moon (up in daytime), the Moon makes no light of its own, and Earth never stops spinning.

33. (A) The Sun lights one full hemisphere of the Moon at all times; phases arise because, from our angle, we see varying amounts of that always-lit half — none at new Moon, all at full Moon, and parts in between. It is not true that only the Earth-facing side gets sunlight, the far half is never lit by the Moon itself, and the Moon never makes its own light.

34. (B) Because the Moon rotates once per orbit, the same near side always points at Earth and the far side always points away — so from Earth we can never see it, though a spacecraft can fly around to view it. The far side is not always dark (it gets sunlight as the Moon rotates), it is not inside Earth's shadow, and it does not permanently face away from the Sun.

35. (C) Day and night anywhere come from how fast that body rotates relative to the Sun. The Moon rotates about once a month, so sunlight stays on one spot for roughly two weeks

and then darkness for two weeks — a day–night cycle of about a month. It is not 24 hours (that is Earth's spin), not an hour, and not a year.

36. (B) Earthshine is sunlight that first bounces off Earth, lands on the Moon's night side, and bounces back to us — a double reflection that faintly lights the dark part. The Moon makes no light of its own, it is not glowing from interior heat, and starlight is far too weak to produce this effect.

37. (B) Day and night come from rotation; spin half as fast and each daylight period and each night period would last about twice as long. The length of the year depends on Earth's orbit, not its spin, so it would be unchanged; seasons come from the axial tilt, not the spin rate; and the Sun would still rise in the east as long as Earth still turns west-to-east — just more slowly.

38. (A) Gibbous means we see more than half but not all of the lit hemisphere; geometrically the Moon is past first quarter but not yet at full, so most of the sunlit side faces us. A Moon directly between Earth and Sun is new (lit side hidden), the new-to-crescent positions show only a sliver, and Earth's shadow relates to eclipses, not ordinary phases.

39. (A) The Sun generates its own light and heat; the Moon makes none and merely reflects a small part of the sunlight it receives, so moonlight is much fainter and not warming. The Moon does not make light independently, moonlight is far weaker than sunlight (reflection loses most of the energy), and it is not stored heat re-emitted at night.

40. (D) Phases depend on the angle between the Sun, Moon, and observer. Even with the Moon fixed, Earth keeps moving (rotating and orbiting), so that angle would still change and we would still see the lit fraction vary — phases would continue. The Moon would not lock to one phase, it would keep reflecting sunlight, and it could never begin to make light of its own.

41. (C) Distance is the key: the Sun is unimaginably nearer than any other star, so it spreads into a disc and floods us with light, while even larger stars are so far away they shrink to points. The other stars are genuine stars (not reflective rocks), the Sun is not the largest star, and other stars do make their own light — they are just extremely distant.

42. (A) Because Earth's axis is tilted, in the polar summer the pole leans sunward and stays in the lit hemisphere even as Earth spins, so the Sun does not set for weeks. Earth never stops rotating, the Sun cannot light a whole sphere at once, and the Moon's faint reflected light has nothing to do with continuous polar daylight.

43. (C) The Moon's spin and orbit take about the same time (~27.3 days), so it turns at just the right rate to keep one face pointed at Earth. A much faster spin or a stopped spin would both let us see all sides over a month, and a slower spin than orbit would not keep the face locked either — equality is what produces the single visible face.

44. (C) Air resistance is what normally slows a feather more than a hammer. With no atmosphere on the Moon, there is no air resistance, so gravity alone acts and both objects fall together. The Moon's gravity is weak (about one-sixth of Earth's), it is not zero (they do fall), and magnetism does not pull a feather — the missing factor is air.

45. (C) Over one ~29.5-day cycle the visible lit fraction grows (waxes) from new to full, then shrinks (wanes) from full back to new — a smooth swelling and shrinking. It does not grow the whole way (it must shrink after full), it is not fully lit the whole time, and the change is gradual over days, not a sudden midnight jump.

46. (A) The Sun is a fairly typical star among billions; it dominates our sky only because it is by far the nearest star to us. It is not the smallest or dimmest, it is not a one-of-a-kind object, and it is genuinely a star (a self-luminous ball of gas), not a planet.

47. (B) The only relevant difference is air: Earth's air resists the light paper more than the dense coin, while the Moon's lack of atmosphere removes that resistance so both fall together. The Moon's gravity is weaker, not stronger; it is not zero (objects still fall); and the paper's mass does not change between worlds.

48. (D) The Sun is a star pouring out its own light, so directly is overwhelming; the Moon merely reflects a small portion of sunlight, so moonlight is gentle. The Moon is not secretly brighter, the Sun does not get its light from the Moon, and the Moon does reflect some light (we can see it), so it does not absorb all sunlight.

49. (D) The Sun lights only the hemisphere facing it; Earth's rotation continually moves any location through the sunlit half (day) and the shadowed half (night). The Sun does not orbit us to take turns lighting cities, city lights are unrelated to day and night, and the Moon's shadow is tiny and rare — it cannot give every city its nightly darkness.

50. (C) Order in the cycle: new → first quarter → full → last quarter → new. So last quarter falls after full Moon as the lit fraction shrinks (wanes). It does not come before first quarter, it is not at new Moon (which shows no light), and it has nothing to do with a lunar eclipse — it is an ordinary phase.

51. (B) A meteor glows because it rams through Earth's air and heats up. The Moon has no atmosphere, so incoming rocks do not burn — they hit the ground and dig craters. The Moon's gravity, though weak, still draws objects in; coldness does not stop burning (friction with air does the heating, and there is no air); and the Moon does not reflect solid rocks away.

52. (A) When you turn one way, the scenery appears to slide the opposite way. Earth spins eastward, so sky objects appear to move from east to west and thus rise in the east, set in the west. They do not appear to move west-to-east (that is the spin direction itself), they are not seen as fixed (the sky clearly drifts), and they do not rise in the north.

53. (A) Because the Moon rotates once per orbit, its far side faces the Sun for part of each lunar cycle and is sunlit then — it is the 'far' side (never seen from Earth), not a permanently dark one. The far side does receive sunlight, the Moon does rotate (so it has a far side), and it is not trapped in Earth's shadow.

54. (D) At most full Moons the slightly tilted lunar orbit carries the Moon just above or below Earth's shadow, so no eclipse occurs; only when the alignment is close enough does the Moon enter the shadow. Earth's shadow does reach the Moon's distance (eclipses do happen), lunar eclipses require full Moon (not new), and the Moon never makes its own light.

55. (A) On Earth, air scatters sunlight after sunset, giving a soft twilight; the Moon has no atmosphere to scatter light, so once the Sun is below the horizon, darkness is sudden. The Moon does not have a thick atmosphere, its rotation is slow (long days), and it has no light of its own to switch off.

56. (B) Strictly, the Moon does not 'shine' on its own — it reflects the Sun's light to us, which we perceive as brightness. It does not burn gases (that is what a star does), it does not store and later release light, and its glow is not from internal heat. Reflection of sunlight is the accurate description.

57. (D) The Moon is our nearest neighbour; the Sun is much farther but is still by far the closest star; every other star is enormously farther still. So nearest-to-farthest is Moon, Sun, next-nearest star. The other orders wrongly put the Sun nearer than the Moon, place a distant star before the Sun, or rank the far star before the Sun.

58. (C) Day and night come from a planet turning so different sides face the Sun. If Earth always kept one face sunward, that side would be in endless day and the opposite side in endless night. It would not give normal day–night (that needs the planet to turn relative to the Sun), the Sun still cannot light a whole sphere at once, and being locked in rotation does not stop the orbit.

59. (C) Both crescents look alike on a single night, but tracking them over days separates them: a waxing crescent is heading toward full Moon so its lit part grows, while a waning crescent is heading toward new Moon so its lit part shrinks. Temperature does not distinguish phases, the Moon always only reflects sunlight in every phase, and crescents are visible from across the world, not tied to single poles.

60. (D) Earth's daytime sky is blue because air molecules scatter sunlight in all directions. The Moon has no atmosphere to scatter light, so sunlight travels straight to the ground and the surrounding sky stays black even with the Sun up. The Moon has essentially no atmosphere (not a dark one), it is not in Earth's shadow at lunar noon, and sunlight clearly does reach it (it is noon there).